# Low-Momentum Scalar Correlations at Positive Gradient Flow in SU($N$) Yang–Mills Theory

A lattice program for four-dimensional moments and the planar limit

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**Status.** This is a pre-results research manuscript. The analytic definitions, large-$N$ counting, tree-level benchmark and lattice strategy are developed here, but no production Monte Carlo result is claimed. It is therefore **not yet a submission-ready research paper**.

**Abstract**

We define and analyze the low-momentum expansion of the connected Yang–Mills gradient-flow energy-density correlator in pure SU($N$) gauge theory. At positive flow time $t = c\,t_0$, the flowed scalar density $E(t,x)$ is ultraviolet finite, allowing the continuum observables

$$\mathcal{X}_E(c,N) = t_0^2\,\Pi_E(0;t), \qquad \mathcal{M}_E(c,N) = -\frac{t_0}{12}\Pi'_E(0;t),$$

where $\Pi_E(Q^2;t)$ is the four-dimensional Fourier transform of the connected $E$–$E$ correlator. Their ratio defines a normalization-independent second-moment scale,

$$\mathcal{R}_\xi(c,N) = \frac{-\Pi'_E(0;t)}{t\,\Pi_E(0;t)}.$$

A tree-level continuum calculation gives $\mathcal{R}_\xi = 3$ and

$$\mathcal{M}_E^{\mathrm{LO}}(c,N) = \frac{3\lambda^2}{2048\pi^2 c}\left(1 - \frac{1}{N^2}\right),$$

with $\lambda = Ng^2$. We derive an economical periodic-lattice time-slice estimator for both the susceptibility and the second moment, discuss continuum and finite-volume limits, and formulate a multi-color program for $N = 3, 4, 5, 6$ and preferably 8. The proposed scalar-channel observables are distinct from the equal-time three-dimensional flowed-energy susceptibility previously used to test large-$N$ factorization, and from the recently determined CP-odd topological susceptibility slope. A possible Adler–Zee interpretation is retained only as a secondary, renormalization-scheme-dependent application. Production lattice data are required before any publication claim is made.

## 1 Motivation and scope

The Yang–Mills gradient flow provides gauge-invariant composite fields at positive flow time with especially simple renormalization properties [1, 2]. This makes flowed observables useful not only for scale setting but also for nonperturbative correlation functions. Large-$N$ studies have already exploited this framework to test factorization with high precision for SU($N$) theories with $N = 3, 4, 5, 6, 8$ [3]. In particular, Garcia Vera and Sommer considered an equal-time three-dimensional susceptibility of the flowed action density.

The present proposal targets a different piece of information: the *four-dimensional low-momentum structure* of the connected scalar correlator. The central quantities are its zero-momentum value and its $Q^2$ slope. Coordinate-space moments of this type have a direct lattice implementation. The general methodology has important recent precedent in the topological channel, where the $Q^2 = 0$ slope has been determined in pure SU(3) [7] and, very recently, in the large-$N$ limit [8]. Those works make it essential to state the novelty narrowly: the target here is the CP-even flowed scalar energy-density channel and its continuum multi-$N$ low-momentum structure, not the invention of the coordinate-moment method.

The original motivation for examining the scalar slope came from the Adler–Zee induced-gravity sum rule. However, the zero-flow scalar trace correlator contains ultraviolet/contact contributions and requires a gravitational renormalization convention. The first paper should therefore focus on strictly positive flow time, where the observables below are unambiguous. The induced-gravity question is discussed only after the finite-flow program has been defined.

## 2 Flowed scalar correlator and low-momentum observables

We use the continuum convention

$$S = \frac{1}{4g^2}\int \mathrm{d}^4x\, G^a_{\mu\nu}G^a_{\mu\nu}, \qquad E(t,x) = \frac{1}{4}G^a_{\mu\nu}(t,x)G^a_{\mu\nu}(t,x). \tag{1}$$

At positive flow time define

$$G_E(x;t) = \langle \delta E(t,x)\,\delta E(t,0)\rangle\,, \qquad \delta E(t,x) = E(t,x) - \langle E(t)\rangle, \tag{2}$$

and

$$\Pi_E(Q^2;t) = \int \mathrm{d}^4x\, e^{iQ\cdot x} G_E(x;t). \tag{3}$$

For an isotropic Euclidean vacuum,

$$\Pi_E(Q^2;t) = \Pi_E(0;t) - \frac{Q^2}{8}\int \mathrm{d}^4x\, x^2 G_E(x;t) + O(Q^4). \tag{4}$$

We introduce two dimensionless observables at fixed $c = t/t_0 > 0$:

$$\mathcal{X}_E(c,N) = t_0^2\Pi_E(0;ct_0), \tag{5}$$

$$\mathcal{M}_E(c,N) = \frac{t_0}{96}\int \mathrm{d}^4x\, x^2 G_E(x;ct_0) = -\frac{t_0}{12}\Pi'_E(0;ct_0). \tag{6}$$

The combination

$$\xi^2_{E,2}(t) = \frac{\int \mathrm{d}^4x\, x^2 G_E(x;t)}{8\int \mathrm{d}^4x\, G_E(x;t)} = -\frac{\Pi'_E(0;t)}{\Pi_E(0;t)} \tag{7}$$

is a four-dimensional second-moment length. Because the flowed operator is smeared in Euclidean space, we use this term as a moment definition rather than identifying it directly with a transfer-matrix correlation length. A particularly clean shape observable is

$$\boxed{\mathcal{R}_\xi(c,N) = \frac{\xi^2_{E,2}(ct_0)}{ct_0} = \frac{12\mathcal{M}_E(c,N)}{c\,\mathcal{X}_E(c,N)}.} \tag{8}$$

It is independent of the overall normalization of $E$ and is therefore attractive for cross-code and multi-$N$ comparisons.

## 3 Large-$N$ scaling and the reference scale

Following Ref. [3], a convenient $N$-dependent definition of $t_0$ is

$$t_0^2\langle E(t_0)\rangle = 0.1125\,\frac{N^2-1}{N}, \tag{9}$$

which reduces to the standard 0.3 condition for $N = 3$ and keeps a fixed 't Hooft-coupling scale at leading order.

In the convention above, $E \sim O(N)$ while the connected two-point function is $O(N^0)$ at fixed $\lambda = Ng^2$. Consequently,

$$\mathcal{X}_E(c, N) = \mathcal{X}_E^{(\infty)}(c) + \frac{x_1(c)}{N^2} + O(N^{-4}), \tag{10}$$

$$\mathcal{M}_E(c, N) = \mathcal{M}_E^{(\infty)}(c) + \frac{m_1(c)}{N^2} + O(N^{-4}), \tag{11}$$

$$\mathcal{R}_\xi(c, N) = \mathcal{R}_\xi^{(\infty)}(c) + \frac{r_1(c)}{N^2} + O(N^{-4}). \tag{12}$$

No extra division of $\mathcal{M}_E$ by $N^2$ is appropriate. This point is important because normalized factorization observables can scale differently from the connected correlator itself.

# 4 Tree-level continuum benchmark

At tree level in Feynman gauge, the flowed propagator contains $g^2 e^{-(t+s)p^2}/p^2$. For two equal-flow energy-density insertions, use $p = \ell + q/2$ and $k = q/2 - \ell$. The quadratic vertex contraction reduces in four dimensions to

$$R(p, k) = 1 + 2\frac{(p \cdot k)^2}{p^2 k^2}. \tag{13}$$

Expanding the resulting one-loop momentum integral to $O(q^2)$ gives

$$\Pi_E(0; t) = \frac{3(N^2 - 1)g^4}{512\pi^2 t^2}, \tag{14}$$

$$\Pi'_E(0; t) = -\frac{9(N^2 - 1)g^4}{512\pi^2 t}. \tag{15}$$

The corresponding one-point function is the standard

$$\langle E(t)\rangle_{\mathrm{LO}} = \frac{3(N^2 - 1)g^2}{128\pi^2 t^2}. \tag{16}$$

Therefore

$$\mathcal{X}_E^{\mathrm{LO}}(c, N) = \frac{3\lambda^2}{512\pi^2 c^2}\left(1 - \frac{1}{N^2}\right), \tag{17}$$

$$\mathcal{M}_E^{\mathrm{LO}}(c, N) = \frac{3\lambda^2}{2048\pi^2 c}\left(1 - \frac{1}{N^2}\right), \tag{18}$$

$$\mathcal{R}_\xi^{\mathrm{LO}}(c, N) = \boxed{3}. \tag{19}$$

The last result is especially useful: the trivial coupling and group normalization cancel, leaving a parameter-free leading benchmark for the low-momentum shape.

A second useful normalization uses the gradient-flow 't Hooft coupling

$$\lambda_{\mathrm{GF}}(t) = \frac{128\pi^2 N}{3(N^2 - 1)} t^2 \langle E(t)\rangle, \tag{20}$$

which equals $\lambda$ at leading order. Thus

$$\mathcal{T}_M(c, N) = \frac{2048\pi^2 c}{3(1 - 1/N^2)} \frac{\mathcal{M}_E(c, N)}{\lambda_{\mathrm{GF}}^2(ct_0)} \tag{21}$$

has $\mathcal{T}_M = 1 + O(\lambda)$ and provides a tree-normalized diagnostic. A precision interpretation requires a higher-order matching calculation.

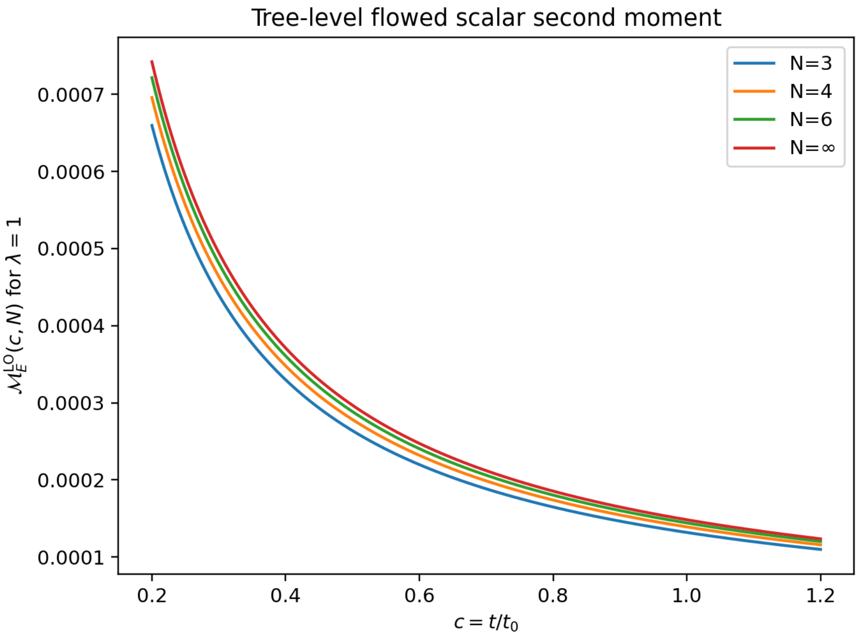


Figure 1: Tree-level $\mathcal{M}_E(c, N)$ for $\lambda = 1$. The planar limit is finite and the leading group dependence is only $1 - 1/N^2$.

## 5 Lattice time-slice estimator

Let $E_L(n)$ denote a dimensionless lattice discretization corresponding to $a^4 E(x)$, and let

$$Y(n_0, t) = \sum_{\boldsymbol{n}} \delta E_L(n_0, \boldsymbol{n}; t) \tag{22}$$

be the spatial slice sum. For a periodic isotropic $L^4$ lattice define

$$C_Y(\tau; t) = \frac{1}{L^3 L} \sum_{n_0} \langle Y(n_0 + \tau, t) Y(n_0, t) \rangle . \tag{23}$$

Then

$$\sum_{\tau} C_Y(\tau; t) = \sum_{n} \langle \delta E_L(n; t) \delta E_L(0; t) \rangle. \tag{24}$$

Hypercubic symmetry also implies, in ensemble expectation,

$$\sum_{n} d^2(n, 0) G_L(n) = 4 \sum_{\tau} d^2(\tau, 0) C_Y(\tau). \tag{25}$$

Consequently

$$\boxed{\mathcal{X}_E(c, N)} = \left(\frac{t_0}{a^2}\right)^2 \sum_{\tau} C_Y(\tau; t), \tag{26}$$

$$\boxed{\mathcal{M}_E(c, N)} = \frac{t_0/a^2}{24} \sum_{\tau} d^2(\tau, 0) C_Y(\tau; t), \tag{27}$$

$$\boxed{\mathcal{R}_\xi(c, N)} = \frac{\sum_\tau d^2(\tau, 0) C_Y(\tau; t)}{2(t/a^2) \sum_\tau C_Y(\tau; t)}. \tag{28}$$

The last ratio is independent of the normalization convention used for $E_L$. The time-slice reduction lowers storage and analysis cost substantially compared with retaining the full four-dimensional density field. It relies on periodic isotropic geometry; open temporal boundaries require a separate bulk-window derivation.

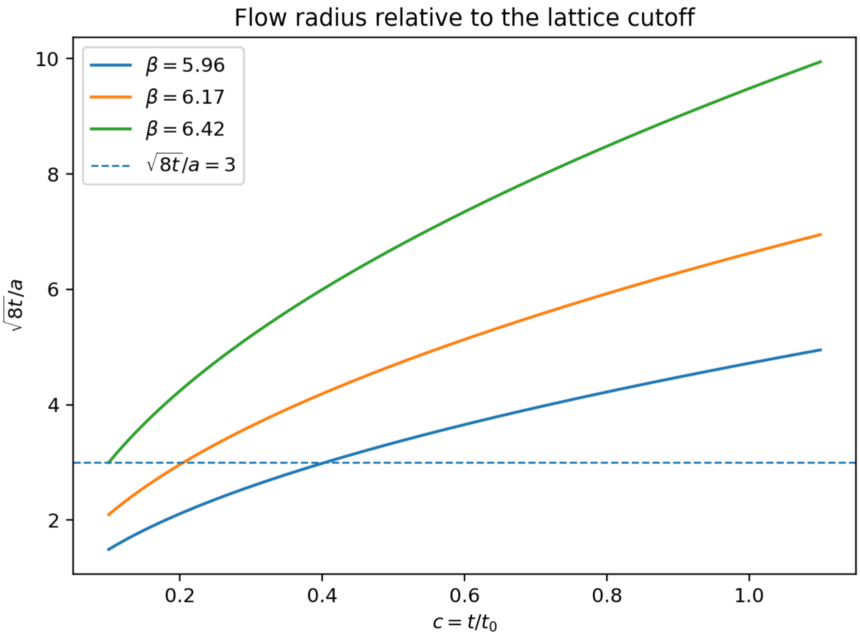


Figure 2: Cutoff diagnostic using reference $t_0/a^2$ values from standard pure-SU(3) scale-setting ensembles. The values are planning benchmarks; the production ensembles must determine $t_0$ independently.

## 6 Continuum strategy and flow-time window

The continuum limit must be taken at fixed positive $c = t/t_0$. A Symanzik-inspired fit has the form

$$\mathcal{O}(c, N, a) = \mathcal{O}(c, N) + k_1(c, N)\frac{a^2}{t_0} + \cdots \,, \tag{29}$$

for $\mathcal{O} \in \{\mathcal{X}_E, \mathcal{M}_E, \mathcal{R}_\xi\}$. The flow radius should be separated from the cutoff. A useful diagnostic is

$$\rho_a = \frac{\sqrt{8t}}{a} = \sqrt{8c\, t_0/a^2}. \tag{30}$$

The first feasibility study can use $c = 0.6, 0.8, 1.0$, but a final publication should include smaller physical smearing where sufficiently fine lattices permit $\rho_a \gtrsim 3$ and stable continuum fits. The analysis should explicitly repeat fits after dropping the coarsest spacing and should compare at least two discretizations or tree-level-improvement prescriptions where feasible [6].

## 7 Statistics, long-distance noise and topology

The $x^2$ weight makes the second moment particularly sensitive to the long-distance tail. Brute-force statistics should therefore not be the only strategy. A multilevel method specifically designed for gradient-flow observables has already demonstrated substantial variance reduction for energy-density correlators [4]. Incorporating that method, or a validated modern equivalent, would materially strengthen the calculation.

The Markov-chain analysis must propagate autocorrelation through the complete estimator. In each blocked-bootstrap replica one should recompute $t_0$, the flow-time interpolation, the slice correlator, the moment, and the continuum intercept. For large $N$ and fine lattice spacing, topological freezing must be monitored even though the target operator is CP even. Recent parallel-tempering-on-boundary-conditions studies show a viable way to mitigate freezing in large-$N$ Yang–Mills simulations [9, 8].

## 8 Relation to existing observables and novelty boundary

Ref. [3] already studied a flowed-energy susceptibility $H_E(c)$ in $\mathrm{SU}(N)$ with $N = 3, 4, 5, 6, 8$. Their integral is equal-time, three-dimensional and unweighted in distance. In contrast, $\mathcal{X}_E$ is the full four-dimensional zero-momentum susceptibility and $\mathcal{M}_E$ is its four-dimensional $x^2$ moment. Together they determine the first two terms of the low-$Q^2$ expansion and the normalized second-moment shape $\mathcal{R}_\xi$.

Coordinate $x^2$ moments are not new in lattice gauge theory. Bonanno determined the topological susceptibility slope in SU(3) [7] and in 2026 extended the calculation to the large-$N$ limit [8]. Those results are in the CP-odd topological-density channel and have direct phenomenological motivation through the proton-spin problem. They sharply narrow the priority claim for the present project. A defensible novelty statement, if supported by complete data, would be:

> First continuum multi-color determination of the four-dimensional low-momentum coefficients of the positive-flow scalar action-density correlator, including its normalized second-moment scale and planar $1/N^2$ limit.

This claim must be rechecked against INSPIRE, arXiv and citation chains immediately before submission.

## 9 Secondary connection to the trace anomaly and Adler–Zee

In pure Yang–Mills theory the trace of the energy-momentum tensor is proportional to the scalar gluonic operator through the trace anomaly. The Adler–Zee formula relates a weighted trace–trace correlator to a matter contribution to the Einstein–Hilbert term. Donoghue and Menezes evaluated the corresponding QCD-like quantity using perturbation theory, the operator-product expansion and lattice glueball information [10].

The positive-flow observables defined here do not require a gravitational counterterm and should be kept conceptually separate from an absolute induced Newton constant. Taking the scalar slope to zero flow reintroduces ultraviolet/contact contributions, including a local-curvature renormalization ambiguity. A later calculation may define a scheme-dependent matter threshold by subtracting a perturbative flowed correlator in a declared scheme, but this should not be the primary claim of the first lattice paper.

## 10 Required numerical results before submission

The present manuscript intentionally contains no fabricated Monte Carlo data. A journal submission should only be prepared after all of the following are available:

1. At least three lattice spacings for each of $N = 3, 4, 5, 6$, and preferably $N = 8$.
2. Stable continuum values for $\mathcal{X}_E$, $\mathcal{M}_E$ and $\mathcal{R}_\xi$ at several common $c$ values.
3. At least one controlled finite-volume enlargement and explicit flow-integrator/discretization tests.
4. Autocorrelation-aware bootstrap errors and a demonstrated treatment of the $x^2$-weighted long-distance tail, preferably with multilevel variance reduction.
5. A meaningful $1/N^2$ planar fit, with an $N^{-4}$ stability test if enough colors are available.
6. A final priority search showing that the scalar-channel low-momentum coefficients have not appeared independently.

## 11 Conclusions

The low-momentum expansion of the positive-flow scalar energy-density correlator offers a mathematically clean and computationally accessible set of SU($N$) observables. The susceptibility $\mathcal{X}_E$, the second moment $\mathcal{M}_E$, and the normalization-independent shape parameter $\mathcal{R}_\xi$ have unambiguous continuum limits at fixed positive flow time, controlled large-$N$ expansions, and simple tree-level benchmarks. The proposal is technically feasible with existing gradient-flow and multilevel methods. Its publication value, however, depends on obtaining genuine continuum multi-$N$ data; the analytic construction alone is not sufficient for a strong research paper.

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